These two points, A and B, will be the endpoints of another segment, AB, on which we choose a point P_t such that $AP_t/AB=t$.

If $t==0$, then $P_t==A$, while when $t==1$ then $P_t==B$. The collection of all these points P_t, for all values of t between 0 and 1, makes a quadratic Bézier curve between A and B.

Notice how the segment A_tB_t is always tangent to the curve.

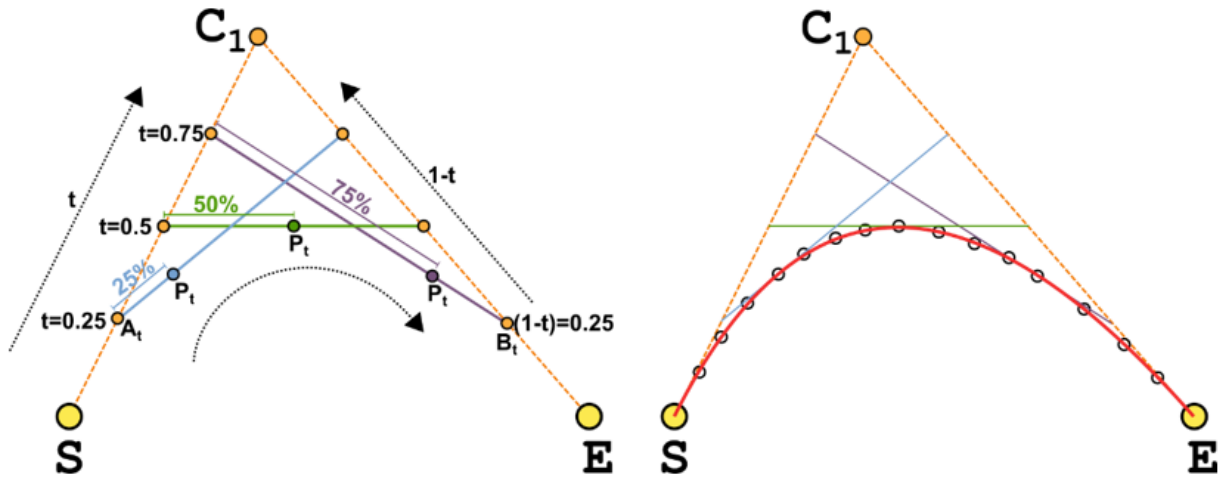


Figure 15.21 How a quadratic Bézier curve is drawn. A quadratic curve requires three points: two endpoints (S, E) and a control point (C_1, or just C for the sake of clarity here); first we have to draw the two segments between the endpoints and C; then, we can draw segments whose endpoints are on SC and EC, under the constraint that if the endpoint on the former, call it A is such that $SA/SC=t$, then B, the remaining endpoint, must satisfy $EB/EC=(1-t)$.
Finally, on segment AB, we choose a point P such that $AP/AB=t$.